

Operations research project

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October 2026

0.1 Mixed-Integer Programming Formulation

The T-ASR problem can be formulated as a Mixed-Integer Linear Programming problem as follows.

Let

$$L = \{\lambda(a, t) \mid a \in A, t \in T\},$$

where $\lambda(a, t)$ denotes the load of arc a at time step t .

The objective is to minimize the arc loads in decreasing lexicographic order:

$$\text{lexmin}(\lambda'(1), \lambda'(2), \dots, \lambda'(|L|)),$$

where

$$\lambda'(1) \geq \lambda'(2) \geq \dots \geq \lambda'(|L|)$$

are the elements of L sorted in decreasing order.

Subject to the following constraints:

Flow conservation

$$\sum_{j \in V \setminus \{i\}} x_{ij}^{d,t} - \sum_{j \in V \setminus \{i\}} x_{ji}^{d,t} = \begin{cases} 1, & i = s_d, \\ -1, & i = t_d, \\ 0, & \text{otherwise}, \end{cases} \quad \forall d \in D, t \in T, i \in V.$$

Maximum number of segments

$$\sum_{i,j \in V, i \neq j} x_{ij}^{d,t} \leq \text{maxSeg}, \quad \forall d \in D, t \in T.$$

Link-load constraints

$$\sum_{d \in D} \sum_{i,j \in V, i \neq j} r^t(i, j, a) v(d, t) x_{ij}^{d,t} \leq \lambda(a, t) c(a),$$

$$\forall t \in T, \quad \forall a \in A \setminus q(t).$$

Budget constraints The original budget constraint contains an absolute value:

$$\sum_{d \in D} \sum_{i,j \in V, i \neq j} \left| x_{ij}^{d,t} - x_{ij}^{d,t-1} \right| \leq \kappa(t), \quad \forall t \in T^*.$$

To obtain a linear MIP formulation, introduce auxiliary variables

$$z_{ij}^{d,t} \geq 0.$$

The absolute-value expression is linearized using

$$z_{ij}^{d,t} \geq x_{ij}^{d,t} - x_{ij}^{d,t-1},$$

$$z_{ij}^{d,t} \geq x_{ij}^{d,t-1} - x_{ij}^{d,t},$$

for all $d \in D, t \in T^*, i, j \in V, i \neq j$.

The budget constraint then becomes

$$\sum_{d \in D} \sum_{i,j \in V, i \neq j} z_{ij}^{d,t} \leq \kappa(t), \quad \forall t \in T^*.$$

Variable domains The routing decision variables are binary:

$$x_{ij}^{d,t} \in \{0, 1\}, \quad \forall d \in D, \forall t \in T, \forall i \in V, \forall j \in V \setminus \{i\}.$$

The load variables are non-negative:

$$\lambda(a, t) \geq 0, \quad \forall a \in A, \forall t \in T.$$

The auxiliary variables used for the budget constraint satisfy

$$z_{ij}^{d,t} \geq 0.$$

Since this is an NP-hard problem AND contains binary decision variables, a branch-and-bound algorithm can be used to solve the MIP. The method divides the feasible solution space into smaller subproblems by fixing selected binary variables to either 0 or 1. For each subproblem, the binary restrictions are temporarily relaxed so that the resulting linear programming relaxation can be solved and used to compute a bound. Branches that are infeasible or cannot improve the best feasible solution found so far are discarded (pruned). This process continues until the optimal integer solution is found.

The program saves time by pruning. This means that the problem completely discards the subtree if there exists one of the following, infeasibility, Suboptimality and when we reach an optimal solution.

The lecturer said instead of trying to make the optimization model decide every individual physical link of a route, we should first look at pairs of nodes.

For each ordered pair of nodes (i, j) , we can use the given network link weights to determine the shortest-path routing from i to j . In the challenge, a segment is exactly such a pair of successive nodes, and traffic between them follows shortest-path forwarding.

SO WE SHOULD Precompute what happens between every pair of nodes, then let the MIP decide which node-pairs should be selected to connect the source to the destination.

1 Formally

The problem is formulated as an MIP by considering each ordered node pair (i, j) as a possible segment. The shortest-path routing for each pair is computed in advance from the network weights, and binary variables $x_{ij}^{d,t} \in \{0, 1\}$ are used to indicate whether demand d selects segment (i, j) at time t . The selected segments must connect the source of each demand to its destination, while satisfying the routing and network constraints.

1.1 Branch-and-Bound Approach

The branch-and-bound procedure is applied to the binary segment-selection variables

$$x_{ij}^{d,t} \in \{0, 1\}.$$

At the root node, the integrality constraints are relaxed to

$$0 \leq x_{ij}^{d,t} \leq 1,$$

while all routing, load, segment and budget constraints are preserved. The resulting linear programming relaxation is solved to obtain a lower bound on the objective value.

If the relaxed solution is integral, then it represents a feasible MIP solution. Otherwise, a fractional variable

$$0 < x_{ij}^{d,t} < 1$$

is selected for branching. Two child subproblems are generated by imposing

$$x_{ij}^{d,t} = 0$$

and

$$x_{ij}^{d,t} = 1,$$

respectively.

Each child problem is solved through its LP relaxation. A node is pruned if its relaxation is infeasible, if its bound cannot improve the current best feasible solution, or if an integer feasible solution has been found at that node. The procedure continues until no unexplored node can improve the best known integer solution.